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"The Safest Woman Alive"

A Reflection on Interpersonal Safety Technologies for Gendered Violence Protection

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Abstract

We tell a story of a woman getting ready to go for a walk, using a plethora of personal safety technologies designed and reported on by HCI researchers to ensure her own safety against public gendered violence (GV). To reflect on this approach, we elicit the Four Domains of Power, highlighting HCI's over-engagement with interpersonal safety technologies when seeking to intervene in GV. In later parts of the paper, we discuss two main contributions for HCI: (1) The Complexities of Designing for GV as an Interpersonal Problem and (2) Complexities of Designing for Normative Understandings of GV to highlight the potential harm in employing interpersonal technological solutions to socio-political issues. Coming back to the four domains of power, we ultimately argue that HCI-researchers and designers can use the framework to analyze their technological interventions to address GV in more nuanced ways as to not re-produce shortsighted, solutionist, or victim-blaming technologies.

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1 "The safest woman alive": A Prologue

You are going for a walk outside. Maybe it's dark, maybe not. Maybe you want to take a walk in the city, forest or through your University Campus? To do so, you have been told, over and over again, that you need to protect yourself from the lurking harm out there. The worst imaginable: catcalls, assault, abuse or even rape. To protect yourself you have invested money and time in a wide range of gadgets suggested to you through various research and other projects.

You open your closet to start getting dressed and begin with putting on underwear, attaching a body sensor to your bra [1]. With a plethora of features the sensor can protect you by calling five close contacts on your phone and recording a potential situation. The sensor can also recognize if you are forcefully disrobed and sound an alarm before spraying an "anti-libido" fluid onto the perpetrator. Second, you put on trousers and the "smart girl security system" belt. By the touch of a button, the belt sends a silent alarm of your current location to three of your closest contacts as well as the nearest police station. Alongside the silent calls, it also sends out a screaming "siren", warning others around you while omitting an electric shock to the potential attacker [2].

After putting on your underwear, trousers and shirt, you clip the smart foot device onto your shoes. If you end up in a dangerous situation, you can activate the foot device by tapping one foot behind the other. By doing so, four chosen contacts will be alerted about your whereabouts [3].

Being fully clothed, you now turn to accessories: The first wristband or "Safeband" in your collection contains a button which, similarly to your belt, sends out coordinates to nearest police station and close phone contacts if pressed. At the same time, a light on your wristband switches on as soon as police are dispatched – if forcefully removed by your attacker, police is contacted once again [4]. You put the "safety system watch" onto your second arm and place the "neck chain pendant" [5] onto your necklace. When in danger, you know you can tap the pendant three times to send out your GPS-location to your close contacts while turning on a video which is sent to "police control centers" - you're not entirely sure what this means or exactly where your data is being sent, stored, and what it's then used for. By calling out a "wake-up-word" of choice the linked watch emits a buzzer noise to ward of the potential predator. You know that if the alarm alone doesn't fend off the attacker, the watch doubles as a tazer, shocking anyone trying to touch you [5]. For additional measure, you add your third "smart bracelet" onto one of your wrists – this one automatically senses, detects and identifies physical abuse. If it detects assault, the bracelet triggers your phone to call the emergency services while sending your location to them [6]. You now have three wrist bands on your arms, but as you don't want to take any chances, you keep adding the safety technologies to your body. The fourth wristband is the "wearable women safety device" which monitors your body parameters such as heart rate and temperature – the bracelet sends this information on to law enforcement and your family along with your location if you activate the system through the touch of a button [7]. You put on your fifth and last wristband from your security closet which through the touch of a button determines your location, sends an alarm message and your location to your close contacts, and your body temperature and heart rate to the nearest police station [8]. Your final accessory is the "Smarisa" ring, which sends the location of any potential attack to your close contacts while filming the attacker [9]. Finally, you put the "women security system" into your handbag: a pepper spray and GPS-location stick with a secret camera that can film any attack [10].

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The last thing you do before you leave the house is to double-check if the security apps you've downloaded onto your smartphone are working. You start by checking the first app called "Safe street" which registers street harassment in real-time and gives you suggestions for safer routes [11]. Then you turn on your second app, which can suggest safe routes and show you which routes are most unsafe at that time when traveling out in public. You also have a panic button in this app which sends a message to your trusted circle and starts an audio recording of any abuse for evidential purposes [12]. Your third app can also send an SOS message to your close friends and contacts while also sensing if someone in your surroundings is carrying a hidden camera and recording audio and video of a potential situation [13]. The fourth app contains lots of valuable information in case of various attacks such as the location of hospitals and police stations and counselling. It can also monitor your health to the point where it sends you a notification if you need to go to the doctor so to prevent unwanted pregnancies after an instance of rape. Moreover, during an attack, by pressing the volume button on your phone, the app sends three calls and texts to chosen contacts and the nearest police station [14]. The fifth app called SafeRoute can in real-time suggest the safest route for you when walking our biking [15]. The sixth app is activated by the push of a button where it sends your current location to close contacts every five minutes. By doing this your location is updated if you were moved around during an attack [16].

If you find yourself in need of a police station or women facility locator, you can find this in your eight app which also offers an emergency button and the potential to be accompanied home by a "scouter" [17]. You have three more apps that also allow you to document your experiences of harassment to warn others or to collect as evidence for a court case: Harassmap [18], Hollaback! [19] and Protibadi [20]. On all three app-websites, you can see a map of where previous incidents have happened.

2 Paper overview

In this paper we do two key things: (1) we reflect on and contextualise how many safety technology interventions in HCI seem to be designed for potential victim-survivors who are encouraged to change their behaviours and activities in public, and (2) generate conceptual language for how we can shift our attention in HCI to addressing the underlying cause of the violence: perpetrators, potential perpetrators and the underlying structural issues of GV. This work sits alongside a growing body of critical research on violence and safety, as well as unsafety, in our post-digital world – where we understand that our physical and digital lives are deeply intertwined [21], [22], [23], [24], [25]. As HCI moves to tackle and intervene in complex socio-political challenges of GV, we seek to direct attention to two complexities which we argue is often overlooked when designing for interpersonal GV prevention: (1) The Complexities of Designing for GV as an Interpersonal Problem and (2) Complexities of designing for normative understandings of GV.

We use this alt-chi format as a platform to open up discussions on moving away from misaligned, shortsighted, techno-solutionist and victim-survivor-changing ideas that may be re-produced when suggesting more technologies for women and gender non-conforming individuals to protect themselves from GV. By doing this, we add to the growing body of work in HCI which demands socio-political and systematic change, not interpersonal technological interventions [22], [23], [24], [26], [27], [28], [29], [30], [31]. We believe the design of digital technologies can contribute to all aspects of this in creative and meaningful ways along a spectrum of designing technologies without advocating for technological solutionism.

3 Designing for Gendered Violence Prevention

Gendered violence (GV), such as sexual harassment, rape, forced marriage, female genital mutilation, physical, economical and psychological violence, are unfortunately experienced by a third of women globally [32]. In this paper we use the term *gendered violence* instead of violence against women and girls (VAWG). While these terms are often used interchangeably, we use gendered violence so as to be inclusive beyond the usually heteronormative discussions in VAWG literatures and support services, including queer women and gender-nonconforming individuals [33]. GV expresses itself through many levels in society and not only through interpersonal interactions: e.g. Anti-LGBT+ rhetoric, abortion bans, and the seeming inability for police forces and governments to prosecute rapists at scale, show some of the breadth and different kinds of violence that can be experienced beyond every-day interactions [34], [35], [36].

It is important to highlight, that even relatively 'small' aspects of GV (e.g. Microaggressions) feed into this wider ecosystem of harm, abuse, and violence (e.g. Systemic injustices and laws that result in refusal of healthcare of women due to their pregnancies) [37]. Issues of GV are closely linked to issues of white hetero patriarchy, meaning that people who don't conform to traditional gender expression, people of colour, those who are disabled or neurodiverse, or have additional support needs are more at risk of violence [21], [38], [39]. Furthermore, depending on stereotypes associated with certain behaviours, victims of violence can be seen as 'bad' victims by formal actors in criminal systems (eg. Police, courts, etc.) as well as in less formalised systems of post-traumatic support (eg. Support services, counsellors, etc.) [35]: for example, it is not rare that a woman's choice in clothing is cited in criminal rape proceedings [40] or that sex workers' harm and/or violence are unfairly dismissed as a 'risk' of their work [41].

On top of this, taboo and prejudice surrounding GV often create complex situations and feelings of shame for many people experiencing harm and abuse, as well as those seeking justice [42], [43]. This shame can also be fueled by stigma surrounding experiences of wi and victim-blaming rhetoric which shifts guilt and blame towards victim-survivors and away from abusers [43]. The deeply entrenched issues of GV are complex and despite both structural and interpersonal preventative actions taken by governments, aid organizations, activists, and researchers around the world, many people continue to face violence every day. HCI researchers are of course also part of this ongoing global work to eliminate GV and abuse. As highlighted in our opening story, a lot of research related to physical safety in HCI centers street harassment or other forms of violence perpetrated by strangers in public spaces. This kind of violence is hyper-visible making it relatively 'easy' to target for HCI researchers as a straightforward problem to 'solve'. Interpersonal violence from strangers on the street is something we

can see a relatively simple solution to by suggesting that if women protect themselves with technological gadgets, they will be able to stop an attack. However, that is not how the majority of violence perpetrated against women and other gender non-conforming individuals takes place [25], [44], [45], [46]. While street harassment is a major concern for many around the world, the reality of violence perpetrated against women and gender non-conforming individuals is often more hidden and complex as the most common form of GV occurs from intimate partners [47] and is perpetrated by people close to victim-survivors [48], [49].

Despite the good intentions behind the technologies centered through our prologue, we add to the arguments of feminist researchers, especially those who focus on topics of violence [27], [39], [50], [51], [52] that we must move beyond designing almost exclusively personal safety devices for potential victims of what seem to be random acts of violence from strangers. Interpersonal violence is part of wider systems of harm, oppression, and violence [50], [52]: experiences of harm are part of patriarchy. Using this terminology immediately places us in a feminist frame, which we use unapologetically in this paper – as many other researchers have done in HCI in recent years [27], [53], [54], [55], [56]. Looking towards feminist framework would give us space to explore wider causes and would also use HCI and design research to understand, address, and explore the underlying root causes, tackling harms where they come from rather than where they are experienced. This includes designing for the people enacting the harm [57].

Like others, we argue that we should discuss the complexities of socio-political structures that uphold GV, and how HCI research (processes and outcomes) can support structural change and interpersonal behaviors that could genuinely reduce experiences of violence. In recent years, HCI researchers have started to engage with wider theories of violence that is perpetrated through white hetero-patriarchy and has begun to unpick the structural issues associated with GV and harm [21], [24], [58], [59]. Often, this work is critical of quick fixes but may still present tangible implications for the design of future technologies. In this frame, we build on this existing critical research related to GV and elicit Patricia Hill Collins' four domains of power as a useful framework for exploring the roles HCI can play when addressing such topics

3.1 The Four Domains of Power

The matrix of domination and four domains of power were originally articulated by Patricia Hill Collins in her book Black Feminist Thought [60]. Initially it was shaped to address the complex marginalization and discrimination against black women in the U.S. but has since been used by many others to explore intersecting experiences of harm and violence [56], [61], [62], [63], [64]. The four domains of power are a framework describing different social levels of oppression and how it is managed through, for example state laws, bureaucracy, cultural norms or everyday interactions. The four domains relate to experiences of (1) structural, (2) disciplinary, (3) hegemonic, (4) interpersonal harm and/or oppression. We adopt this framework to discuss how interpersonal safety technologies designed to stop GV often falls short in addressing higher levels of power-domains. We discuss the four domains of power for two main purposes (1) to highlight how GV and patriarchal oppression takes different forms and is organized on different levels in society and (2) to highlight how the examples we position as part of our prologue primarily fall into the interpersonal domain; building our argument for the need to expand HCI research about digital technologies and reduction of GV into the three other domains of power.

The most abstract, largest and static domain is called the **structural domain**. This entails how oppression (in our case GV) is organized through large scale social institutions such as the educational and legal system and through economic imbalance. As a result of patriarchal oppression through the structural domain, we note there is less funding and opportunities for girls and women to go to school [65], and structural inequalities related to funding for women shelters and aid-organizations [66], as well as women's health initiatives [67] – of course these systematic issues are compounded by additional axes of marginalization such as race, ethnicity, disability, neurodiversity, immigration status, and so on [39], [56], [63]. Moreover, we note how many laws and regulations regarding women's safety are not sufficient for their protection from GV and patriarchal oppression [68]; which we must also directly link to ongoing undermining of women's safety in large western economies. Furthermore, there are even fewer structural protections for trans, queer, and other gender-nonconforming people [69], [70] (or there are active movements to undermine and reduce the protections queer people have fought to have), who must also be included when discussing GV.

The second **disciplinary domain** includes organizational practices and bureaucracy. This disciplinary domain relates to patriarchal oppression through regulations and control. Examples of these practices include the difficulty for many women to press charges against attackers or abusers through a complex and often stigmatizing justice system [34] - as we highlighted above this would also include the hierarchies of victimhood [71] making it ever more difficult for those who are deemed 'less deserving' of justice to argue their case. It also includes the common trivialization of reports on sexual assault and the often victim-blaming processes of testimony [34], [72]. For example in the UK only 1.3% of rape cases documented by police result in a suspect being charged [73]; this inevitably leads to many victim-survivors not seeking legal justice routes.

The third **hegemonic domain** legitimizes patriarchal oppression through the reproduction of harmful norms, cultures and ideologies. Between different countries and cultures women are perceived very different. In more conservative cultures women can be perceived as 'belonging to men' and be responsible for their honor which is then often cemented through laws and the media in such countries [74], [75]. These cultural understandings are of course not universal but are felt differently across cultures and countries. However, the hegemonic domain manages oppression even in the most liberal countries where conservatism might not be as visible but where women are still often shamed for their sexuality [76], [77].

The last and most visible domain is the **interpersonal domain**. For this domain we note individuals direct lived experience and day to day interactions of, for example, physical and psychological GV such as is [15], [17], [48], [59], [78]. These practices also include public sexual harassment and abuse by strangers as well

STRUCTURAL	DISCIPLINARY
Organizes oppression through large-scale social institutions (macro-level)	Manages oppression through practices and bureaucratic rules
HEGEMONIC	**INTERPERSONAL**
Involves ideological and cultural norms. Shapes beliefs, norms and ideas	Everyday interactions and experiences of oppressoin through personal relationships (micro-level)

STRUCTURAL	DISCIPLINARY
Less funding for girls education and opportunities for attending school Little support for shelter and aid organizations Little funding for womens health Laws undermining girls and womens rights	Upholding complex legal structures causing difficulties in reporting abuse The trivialization of women and girls' stories of abuse and assault through
HEGEMONIC	**INTERPERSONAL**
Upholding and reproducing victim- blaming ideas through media Upholding and reproducing stigma and taboo ideas on female sexuality	Direct IPV Public sexual assault and harrassment Female genital mutilation Stalking

Figure 1: The four domains of power a) original model b) examples contextualize towards gendered violence

as direct intimate partner violence which might be considered personal issues but is however highly rooted in the other domains of power rendering GV [60]. Since this domain is highly visible and concrete in its form (in most cases because it is what people can see with their own eyes) the way to counteract interpersonal violence can happen through interpersonal safety technologies in which we report about in the prologue of this paper.

In discussing design for GV prevention and bringing in the four domains of power to contextualize different social levels organizing GV we now move to further examine complexities arising when treating GV as an interpersonal problem and a normative understanding.

3.2 Complexities of Designing for Gendered Violence as an Interpersonal Problem

Despite the many opportunities arising from HCI researchers positioning technology and design as a means of reducing GV and tools for systemic change, many contributions of anti-violence technologies particularly suggest women to change their behaviors and actions in public, which we highlight in our prologue story (See e.g [3], [4], [5], [9], [11], [14], [16]. While interpersonal safety technologies are definitely an important part of widespread violence prevention, we highlight how an exaggerated attention to individual change can foreground misplaced trust in technology and elicit further harm and stigma to those we are trying to help. For example, Sultana and colleagues, (2018) initially went into their field site with the intention of spending time together with women from rural Bangladesh to guide them towards ideas of technologies which could improve their lives. However, the research team quickly realised that the sheer weight of the socio-political issues faced by their participants was *"beyond the scope of design interventions of the scale and forms we are used to in HCI."* We extend the insights from Sultana et al (2018) towards violence-prevention technologies foregrounding how socio-technical interventions need to address the many complex nuances of GV while also demanding a larger attention to socio-political and cultural structures. To contextualize our arguments, we give examples of safety technologies which are thought to help women protect themselves from interpersonal GV, such as smartphone applications that claim to lead women along 'safe' routes and gadgets that send information about attacks to emergency services and close family members.

In suggesting a GPS-based smartphone application where women choose their route to work depending on where reports of street harassments have been made, we recognize the intention to *"facilitate women to feel secured to boost self-esteem and participate in daily-life activities"* [11]. Similarly, in proposing security systems for women faced with critical issues that track and send locations [5], [7], [14], [15], [16], sound alarms [1], [5] and omit electric shocks [2], we also understand and support the intention of wanting to: *"solve the (critical) issues [faced by women] with technologically sound equipment's and ideas"* [2]. However, we must consider the wider impacts of these kinds of suggestions. In cases of safety technologies which suggest alternate routes for women, we draw on research that highlights that the 'safety' of routes is influenced by peoples' intersecting identities [79], [80], [81], and when such types of routing systems were for example, rolled out via Microsoft, they garnered backlash from many, including the American Civil Liberties Union [82], [83]. Such ideas lead us to asking questions related to which routes are 'safe for whom'? We also want to question the implicit notion that women should avoid certain locations to protect themselves – arguably limiting their movement and reducing the space they are allowed to take up in the world.

While many safety gadgets carry features for women to fight back in situations of violence, we argue that a broader analysis of *why* women need to be handed for example apps for location tracking, siren alarms, panic buttons and electric chocks would benefit the HCI research community – and whether they are in line with contemporary discourse related to feminist safety and security research [17], [56], [79] as well as trauma-informed, harm-reduction and violence-prevention HCI research [24], [59], [78][84]. Importantly, this should intersect with developments in intersectional and specific-community needs, such as the inclusion of neurodiverse [85], gender non-conforming [86], trans [69], and fat [87] bodies in our design processes; including specific design research related to safety concerns of these communities [70]. We also argue that

HCI-researchers need to analyse the implications of suggesting that technological safety artefacts will be "*solving issues faced by women"* [2], *"prevent sexual assault"* [1], or "*ensure the safety to the women"* [4] in deeper, more nuanced ways. There have been instances of socio-technical artefacts that have been used to for example bring in important attention to the width and depth of GV in Egypt through the website Harassmap [18] or to reveal the everyday reality of street harassment for many women in New York through the platform Hollaback (now called Right To Be) [19] we continue to highlight that the complexities width of GV as a large structural issue of systemic nature is too big to solemnly fight with technological devices alone. For example, some of the success of Harassmap has come not only from the collection of data, but also from its use for activists, in courts, and so on. These distinctions are important to make so as to not reproduce techno-solutionist arguments.

With this discussion, we add to existing concerns about solutionist rhetoric in HCI [48], [78] which puts the responsibility of violence protection/prevention on women and gender non-conforming individuals as opposed to social institutions feeding into structures of GV. We also argue that by suggesting women to be the responsible party to prevent interpersonal violence by buying, learning to use, use correctly, and limit their lives to correspond with any safety technology, the burden of addressing GV in wrongfully placed.

3.3 Complexities of Designing for Normative Understandings of Gendered Violence

Technology has the potential to change and assist in many processes, and for other purposes than "solving" systemic oppression and GV we recognize its value in for example providing tools of support for victim-survivors [20], [25], [78] or to design *within* the patriarchy in empowering women without putting them at risk for further vulnerability [59]. In doing this we move away from direct techno-solutionism and misaligned safety responsibilities. When exploring potential design "solutions" and interventions to fight GV, it is not uncommon for HCI researchers to shape their work after normative understandings of interpersonal public-violence scenarios such as verbal harassment and physical assault [1], [3], [5], [6].

To address street harassment in public environments some of the examples in our prologue center technological solutions employed in infrastructure so as to be able to automatically sense GV. Through for example an emergency button installed in company cars notifying the company and nearest police station when a woman feels discomfort or an automatic female-fear and anger emotion recognition device for metro systems initiatives aligns towards the protection of female commuters [88], [89]. While we again recognize the well-intended-meaning and sense of urgency in these cases, it is the social contexts in which these solution sits that becomes problematic: we know that facial recognition systems are not trained on diverse-enough bodies [90], [91] and that emotion-recognition is near-impossible [92], resulting in untrustworthy algorithms that may cause more harm than do good. Furthermore, this kind of technology assumes that visible alarms will trigger others to help and discourage an attacker, which is not always the reality [93]. Moreover, we argue that any gender non-conforming individuals would probably be hesitant to interact with any such public infrastructure system since increased structural stigma and taboo could mean that attention from the public, police and authorities could lead to even greater violence [69], [70], [94].

Taking all of the above to its logical conclusion, we argue that by designing safety technologies based on specific normative scenarios of GV, we require victim-survivors to correctly analyze any upcoming dangers, take the correct preventative measures so as to activate the corresponding protective features, all while also fitting the normative profile of 'a woman'. This means that artefacts which are put forward as part of the solution for GV only work if each imagined violence scenario of abuse happens in the intended way, which is highly unlikely. Furthermore, sensor-driven devices must be triggered by specific 'expected' and normative responses to potential harms (even if we know that people react differently to forms of violence based on their previous experience – sometimes referred to fight, fight, or freeze reflexes [95], [96]).

When wanting to design interpersonal safety technologies to reduce GV, it is not wrong to think that a top-of-the-line technological gadget should be able to perform all of the safety measures that we highlight in our prologue or in this section. However, we argue that it becomes rather absurd to think that the safety of women and gender non-conforming individuals should depend on (1) using products correctly or (2) ensuring normative gender presentations and reactions. That products are able to predict and ward off every possible situation of GV, IPV and sexual abuse. This means coming back to one of our main arguments in highlighting how no HCI safety technologies can single handedly solve or prevent GV and how the responsibility of doing this should not fall majorly on victim-survivors.

4 Resisting the urge to design: Implications for research

Starting off this paper with our prologue, we introduced a variety of wearable safety technologies and mobile phone apps meant to protect women from GV scenarios. We then discussed some of the complexities and pitfalls of putting the responsibility of GV prevention and protection on victim-survivors as misaligned and over-emphasized. Secondly, we argued for the shortcomings of designing technological safety devices which are thought to prevent any possible outcome of interpersonal GV if only used correctly by victim-survivors and normative bodies. Coming back to the four domains of power to contextualize how different levels of social practices organizes oppression of GV (see section 3.1), we now highlight and build on the framework as a valuable tool for HCI-designers seeking to address GV and the many issues arising from it. We argue that HCI-researchers and designers alike can use these domains to analyse which of the levels they are designing technological interventions for and addressing GV on. We believe, this reflection can assist designers in avoiding misaligned, shortsighted, solutionist or victim-blaming tendencies of digital technologies which we outlined throughout section 3. Using the framework can better crystalize which level of practice the intended technology is situated in.

STRUCTURAL	DISCIPLINARY	STRUCTURAL	DISCIPLINARY
Organizes oppression through large-scale social institutions (macro-level)	Manages oppression through practices and bureaucratic rules		Yoo & Dourish,. (2021) Ahmed,. et al (2014) Dimond et al., (2013) Grove,. (2025)
HEGEMONIC	**INTERPERSONAL**	**HEGEMONIC**	**INTERPERSONAL**
Involves ideological and cultural norms. Shapes beliefs, norms and ideas	Everyday interactions and experiences of oppressoin through personal relationships (micro-level)		Sarosh,. et al (2016) Monisha,. et al (2016) Lindsay,. et al (2013) Mohan,. et al (2017) Viswanath,. et al (2016) Navya, et al (2018) Gautam,. et al (2022) Mahmud,. et al (2017). Yarrabothu,. et al (2015) Islam,. et al (2018) Lokesh, & Gadgil (2017) Levy,. et al (2020) Patel & Hasan (2018) Navya et al., (2018) Gopalakrishnan,. et al (2023) Chougula,. et al (2014) Ali,. et al (2015). Jatti,. et al (2016)

Figure 2: The publications from our prologue contextualized into the four domains of power

To show how the framework can be used we have added each artefact from the prologue into four domains of power (see image 2). As noticeable in image 2, most artefacts from our prologue are aimed at addressing the interpersonal domain specifically in context of public street harassment. Only four papers which we included in our prologue discuss wider complexities of designing for GV is beyond the specifically designed technology. These papers [17], [18], [19], [20], which do not directly design an artefact and instead report on already existing ones, all discuss how large challenges of systemic nature such as GV demand interventions and attention across all domains and on all levels in society.

The visual documentation of our analysis highlights that there is a disproportionate focus on interpersonal violence in anti-violence technology research in HCI. We do recognize how the structural, administrative and hegemonic domains are harder to design for, understand and address. However, this should not be seen as challenges not fitted for HCI but rather imperative factors to consider and analyze when seeking to address issues of GV. In this analysis, we only include papers that designed or reported on specific wearable and/or digital applications that aim to prevent or protect from violence – we are aware there is much critical research in HCI that discusses GV on a more structural scale. However, this research (which includes some of our own) is often missing practical applications and/or anti-violence intervention.

What we sought to do in this paper is to add to the ongoing discussions of how the many designs available as interventions and solutions to GV for women all around the world often miss opportunities when it comes to substantially addressing structures of patriarchal oppression; something that is inherently difficult, complex, and complicated to do. To help researchers reflect on their design processes as it relates to GV we present four implications together moving forward towards larger framings of GV prevention research in HCI.

4.1 Good Intentions are not enough

We understand that the research examples part of the prologue is all conducted with good intention and a sense of urgency – we do not contest this. We also recognize that there are different global and local community factors that make the need for certain interventions more urgent than others (eg. [20], [59], [97]). These aspects have not gone un-recognized in HCI and are imperative perspectives to analyse [59], [78]. However, despite the well- meaning intentions behind much of the work which we highlighted in this paper, there is a disproportionate focus on interpersonal change and crisis response. This inevitably means that preventative and other protective approaches that center systemic change through socio-technical artefacts are underrepresented. In turn, this can create an over-justification for techno-solutionist rhetoric and technologies, reducing other potentially less lucrative and 'innovative' but more mundanely useful systems.

4.2 Scrutinizing Academic Pitfalls

We would like to extend the discussion around the very structures of our research field and academic career paths. We recognize how a straightforward aesthetically pleasing design that addresses highly visible social issues are probably considered more "attractive" not only to ECR´s who works to make their research visible but also to conferences, journal formats publishers and funders. Having something tangible to show out of research, such as a working gadget, can feel like a more concrete contribution than, suggestions, considerations and reflections of a socio-political nature. In reality though, many of these kinds of 'solutions' rarely make it out of our insular world of HCI; arguably, this is not necessarily inherently problematic, but it becomes so when presented as solutions to complicated problems It is here again, where we want to highlight the need of working in allyship with support services, meaningfully, and explore how our research can function as part of wider prefigurative politics [26] and contribute to ongoing activist efforts [98].

4.3 This problem is bigger than HCI can fix

As we have pointed out throughout the paper, HCI researchers are only one small part of wider work that aims to eradicate GV. In this paper, we are centering the HCI community for a variety

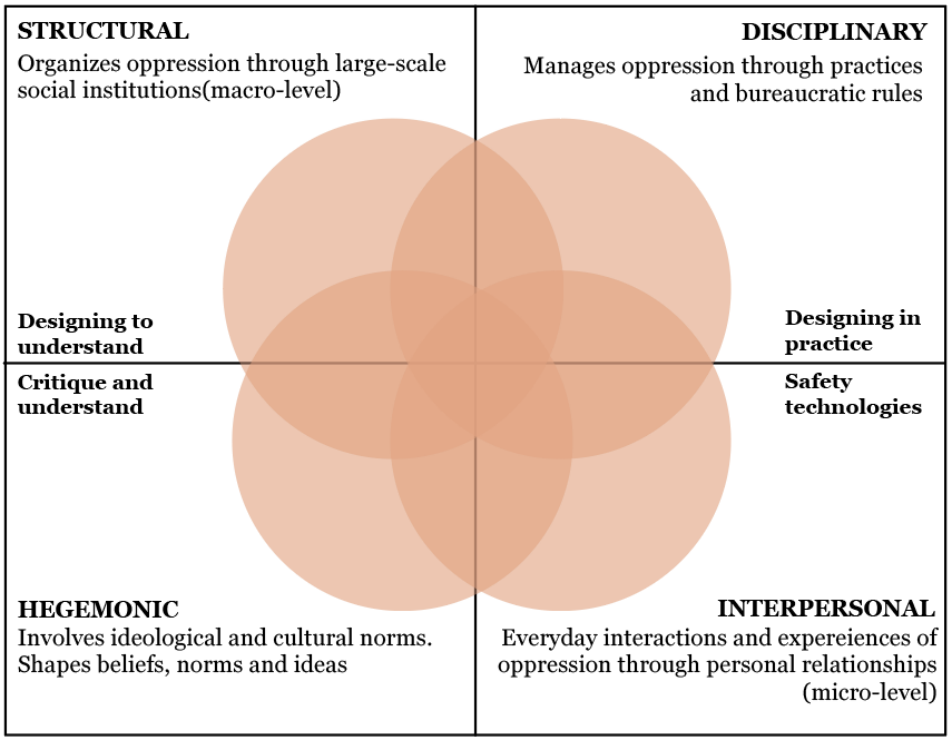


Figure 3: The four domains of power including four circles of potential HCI focus

of reasons: (1) because we are a part of it and (2) to understand that there is work to be done within our own community that can advance understanding of violence prevention and reduction (including work that needs to be done within our own community to reduce violence [99], [100], [101], [102], [103]. But what we are talking about is bigger than HCI – we need to not just address the issue of GV as a digital intervention one, one that is related to research only, or one that only focuses on the very important crisis points. There is some fantastic work that understands research and design in partnership with support services [24], [49], [78], [104], [105], and we need more of this. Importantly though, we do not just need to work 'in partnership' but also in allyship, in affective alliances, in networks of care, and in justice-oriented ecologies that tackle violence at its roots, that provide genuine preventative support, and that continue to intervene in crisis situations (albeit with more nuanced understanding of what might actually work in those situations) – moving forward the discussion on and interventions for the reduction of GV. This is not solely a research concern, but one that is global, international, national, and hyperlocal.

4.4 HCI beyond gadgets and critique: developing a framework of safety-HCI research

We argue that there needs to be a larger conversation in the HCI community to imagine more ambitious futures related to GV. We need new imaginaries that support our re-evaluation of our own position in addressing systemic oppression such as GV. Some researchers have started to ask questions about what a trauma-informed design and/or social media space could look like [84], [106], but we want to go further and ask for alternative forms of harm reduction and violence prevention? What would a design process look like that intervenes in non-carceral ways at early signs of willingness to perpetrate violence? To respond to some of these questions, we present a framework for different trajectories of work that range from technological gadgets that address a very specific anti-violence need (such as the ones we described in our opening story); opportunities for designing in practice – developing tools and resources with and for activists, support organisations [21], [105], [107], [108], [109], [110], and those working to eliminate harm by working with perpetrators; designing to better understand and re-imagine what HCI research might be able to do when we discuss harm reduction approaches related to gendered (and racial) violence; and all the way to the use of HCI research and design processes of digital technologies to critique and understand (see image 3). We believe that HCI and Design research can be useful in the ongoing work to reduce violence across all areas along this trajectory from gadgets to critique – but importantly research must relate to all of these aspects, situated in context of the four domains of power.

5 Conclusions

This paper examines the complexities of designing for interpersonal and seemingly instant responses to longstanding, deeply rooted systemic oppression. To highlight the complexities of designing

for interpersonal GV protection we told a story of a woman getting dressed to go for a walk thinking she needs to wear and use a plethora of technologies to ensure her own safety. We do this to bring attention to what we argue is an over-emphasis on interpersonal violence-prevention and crisis-response produced by HCI researchers. To make our argument, we used the four domains of power as originally articulated by Patricia Hill Collin, to contextualize other equally important social levels that also manages oppression. By discussing HCI papers suggesting and reporting on interpersonal technological artefacts to intervene in GV scenarios, we discussed (1) The Complexities of Designing for GV as an Interpersonal Problem and (2) Complexities of Designing for Normative Understandings of Gendered Violence. We then brought in a discussion on the potential harm in employing interpersonal socio-technical solutions to socio-political issues and come back to the four domains of power as a tool in which we argue that HCI-researchers and designers alike can use to analyse on what level they are designing technological interventions that address GV so as to avoid the simplistic views that lead to techno-solutionism that we see too often. Our contribution expands the ongoing critical wave in HCI that closer examines technology as tools of power. While we do not contest designers' intentions nor design, in this paper we frame both the dangers in only addressing potential victim-survivors in a crisis situation and the opportunities of analyzing the larger socio-political structures causing GV. Ultimately, we present a framework that HCI researchers can use as a reflective tool when designing anti-violence and/or harm reduction approaches.

We hope that this framework will be useful for researchers as a reflective tool to explore where their work is situated. With this knowledge, we are then able to explore how our work relates to the other domains of power and ways of doing HCI research. This is of course not a complete tool and should be read in conjunction with specific literatures related to the specific issues the researchers are aiming to combat (eg. safety for LGBT+ people, black women, neurodivergent, and disabled people, etc.). We also want to contextualize this framework with the understanding that it is unlikely that, as society, we are going to end all violence. Here, we take inspiration from To and colleagues [111], who are re-imagining anti-racist technologies through lenses of joy and playfulness in design: What would an alternative imagination for technologies and their design processes look like, if we took ecosystems of harm and violence into consideration and re-imagined our frame of reference?

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